DDI-Net 3.0

Explainable drug interaction intelligence

Drug–Drug Interaction Research Report

Warfarin + Fluconazole

DRUGS ANALYSED

A **Warfarin**
DB00682 · approved

B **Fluconazole**
DB00196 · approved, investigational

Report ID	DDI-20261015-045145-80DDD4EA
Generated	15 October 2026, 04:51:45 (UTC, UTC+00:00)
Generated (UTC)	2026-10-15T04:51:45.099Z
Model	BIO-GINE M4 · bd45f84e3c1b2c33 · seed 0
API report version	3.0
Analysis produced by the service	2026-10-15T04:51:44+00:00
Response received by the browser	2026-10-15T04:51:44.755Z
Service	http://127.0.0.1:8765
Language	English

CONTENTS

1 Executive summary
2 Drug information
3 Documented interactions
4 Biological mechanisms
5 BIO-GINE research analysis
6 Structured explanation from the service
7 Sources and reproducibility
8 Limitations
A Machine-readable summary (JSON)

RESEARCH USE ONLY

DDI-Net is a research prototype — not a medical device and not clinical decision support. The absence of a documented interaction is not evidence of safety. The BIO-GINE score is a research signal of resemblance to documented pairs, not a probability of clinical harm.

1 Executive summary

Documented in a database

Interaction documented in a database; the source gives no severity

A database describes an interaction; it gives no severity.

This is the kind of documentation found, not a level of danger.

FOUR KINDS OF INFORMATION IN THIS REPORT

Documented evidence

DrugBank records: 1; label excerpts: 0

A source states that these two drugs interact.

Mechanism hypotheses

hypotheses: 3; shared Reactome pathways: 10

Assembled from each drug's documented roles separately; not tested for the pair.

Machine-learning prediction

calibrated score 0.811 · regime S2

BIO-GINE's resemblance to documented pairs; not a probability of harm.

Missing or unavailable evidence

FDA labels (openFDA): unavailable

What could not be checked, or is absent from the sources.

- The pair analysed is Warfarin (DB00682) and Fluconazole (DB00196).
- Kind of documentation found: Documented in a database. This is the kind of source, not a level of danger. DrugBank interaction records: 1; official label excerpts: 0; label mentions reporting no effect: 0; conflicts between records: 0.
- Mechanism hypotheses assembled from documented drug–protein roles: 3 (CYP2C19 — inhibits a protein that handles the other drug; CYP2C9 — inhibits a protein that handles the other drug; CYP3A4 — inhibits a protein that handles the other drug). None was tested for this pair. Shared Reactome pathways: 10 (background, not a cause of interaction).
- BIO-GINE M4: calibrated score 0.811 (raw 0.999972), regime S2. The score expresses resemblance to documented pairs; it is not a probability of clinical harm and carries no severity.

Uncertainty and evidence gaps

- Source “FDA labels (openFDA)”: unavailable. Its data is not part of this report.
- For 1 of 1 DrugBank records the drug order in the export is not verified, so the report does not say which drug affects which.
- The service uses one model checkpoint (seed 0) and provides neither a per-pair confidence interval nor an ensemble estimate. Measured performance for this regime is given instead — an average over many pairs, not confidence in this pair.
- No source gives a severity, and none is assigned here.
- Doses, duration, age, kidney and liver function, other medicines and individual factors are not taken into account.

2 Drug information

FIELD	A · WARFARIN	B · FLUCONAZOLE
International name (INN)	warfarin	fluconazole
Russian name	варфарин (transliteration)	флуконазол (transliteration)
DrugBank ID	DB00682	DB00196
DrugBank groups	approved	approved, investigational
Type	small molecule	small molecule

FIELD	A · WARFARIN	B · FLUCONAZOLE
Query as entered	DB00682	DB00196
Matched on	—	—
Protein annotations (targets / enzymes / transporters / carriers)	2 / 6 / 0 / 2	1 / 4 / 1 / 0
Model split (seed 0)	val	train
Protein data available to the model	yes	yes
DrugBank page	https://go.drugbank.com/drugs/DB00682	https://go.drugbank.com/drugs/DB00196

Identifiers: DrugBank. INN: DrugCentral, joined on InChIKey (no value means no match). Russian names: rule-based transliteration — an interface label, not sourced data.

3 Documented interactions

Each record is numbered; the explanations refer to these numbers. The English text is the source's own words (an excerpt or DrugBank's standard sentence).

[1] Database · DrugBank | direction not verified Retrieved 2026-08-24

SOURCE'S WORDS · DRUGBANK TEMPLATE, DIRECTION-NEUTRAL

“The serum concentration of one of the two drugs can be increased when it is combined with the other drug.”

direction not verified: the report does not say which drug affects which. Order check for this type: 1,270 of 12,308 checkable records consistent (10.3%).

Source	DrugBank DDI records via Therapeutics Data Commons
Version	PyTDC 1.1.15, tdc.multi_pred.DDI(name='DrugBank')
Record ID	DB00682|DB00196|Y=73
Original identifiers	drugbank_id_drug1_as_exported: DB00682 drugbank_id_drug2_as_exported: DB00196 tdc_interaction_type: 73
URL	https://tdcommons.ai/multi_pred_tasks/ddi/
Retrieved	2026-08-24
Licence	TDC: CC BY 4.0; underlying DrugBank content: DrugBank non-commercial licence
Severity as stated by the source	not stated by the source

QUALIFIERS

- DrugBank's standard sentence for this interaction type, not a quotation from a study or a label.
- This export carries no severity grade. None is implied.
- The export does not state the basis of each record (clinical study, label text, or inference from pharmacological properties).
- Drug order for this interaction type matched documented DrugBank roles in only 1,270 of 12,308 checkable records, so the statement is shown without naming which drug is affected.

Sources checked

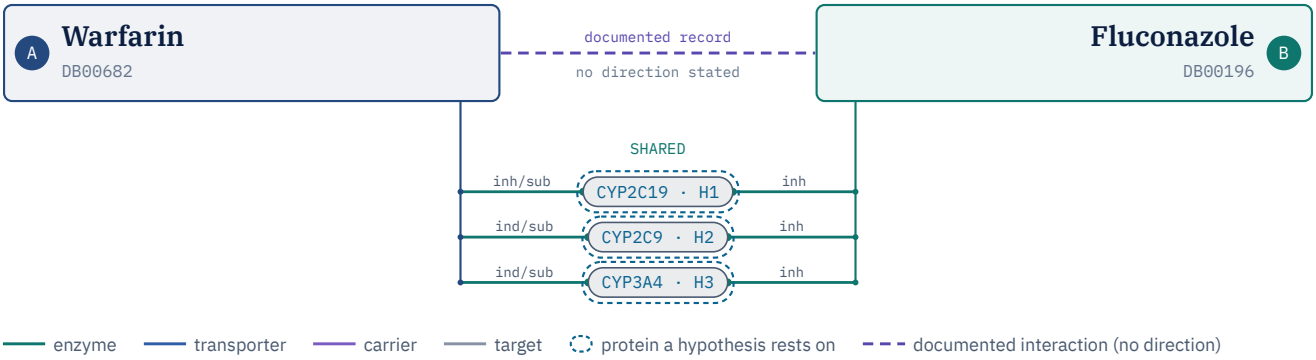
SOURCE	STATUS	DETAIL
DrugBank (TDC export, local)	checked	1 record(s) in the local DrugBank export (TDC, retrieved 2026-08-24).
FDA labels (openFDA)	unavailable	openfda_label: network error (URLError)

4 Biological mechanisms

Below are each drug's documented roles on their own (DrugBank, ChEMBL) and how they line up. These are mechanism hypotheses, not evidence of an interaction: the databases do not record strength of effect or how much of a drug's processing a protein handles.

Mechanism graph

Drawn from the report's mechanism block: proteins documented for both drugs. The full graph, including proteins of only one drug, is on the Mechanisms page.



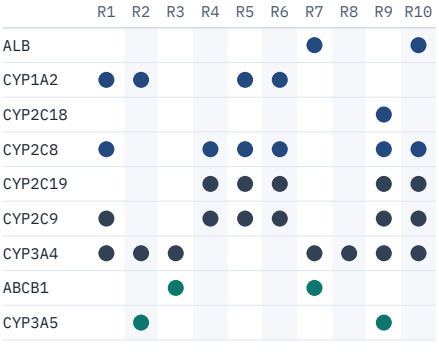
Roles on branches: sub — substrate, inh — inhibitor / inhibits, ind — inducer, act — activates, n/s — not stated.
The arc between the drugs is a documented interaction record. It has no arrowhead: the DrugBank export's drug order is not verified for this type.

Mechanism hypotheses

#	PROTEIN	CONFIGURATION	ROLES A / B	BASIS
H1	CYP2C19 Cytochrome P450 2C19 P33261 · enzyme	inhibits a protein that handles the other drug hypothesis: Fluconazole → Warfarin consistent with record [1] (not independent confirmation)	A: inhibitor, substrate B: inhibitor	DrugBank drug-protein annotation 5.1 (export 2025-01-02) PMID 16141567, 10709776, 19515014, 16162970, 16372822
H2	CYP2C9 Cytochrome P450 2C9 P11712 · enzyme	inhibits a protein that handles the other drug hypothesis: Fluconazole → Warfarin consistent with record [1] (not independent confirmation)	A: inducer, substrate B: inhibitor	DrugBank drug-protein annotation 5.1 (export 2025-01-02) PMID 16141567, 8801056, 10952477, 11475469, 10709776, 29283396, 16372822, 20735727
H3	CYP3A4 Cytochrome P450 3A4 P08684 · enzyme	inhibits a protein that handles the other drug hypothesis: Fluconazole → Warfarin consistent with record [1] (not independent confirmation)	A: inducer, substrate B: inhibitor	DrugBank drug-protein annotation 5.1 (export 2025-01-02) PMID 16141567, 8801056, 15969931, 19515014, 16372822, 20735727

Shared biological pathways (Reactome) · 10

10 of 10 shown: pathways through a shared protein first, then the most specific. Protein membership in Reactome pathways (a dot means the protein is in the pathway; colour shows whose protein it is: A, B or shared).



protein of A · protein of B · shared protein

R	PATHWAY	REACTOME	PROTEINS	VIA A	VIA B
R1	Biosynthesis of maresin-like SPMs	https://reactome.org/content/detail/R-HSA-9027307	6	CYP1A2, CYP2C8, CYP2C9, CYP3A4	CYP2C9, CYP3A4

R	PATHWAY	REACTOME	PROTEINS	VIA A	VIA B
R2	Aflatoxin activation and detoxification	https://reactome.org/content/detail/R-HSA-5423646	8	CYP1A2, CYP3A4	CYP3A4, CYP3A5
R3	Atorvastatin ADME	https://reactome.org/content/detail/R-HSA-9754706	8	CYP3A4	ABCB1, CYP3A4
R4	CYP2E1 reactions	https://reactome.org/content/detail/R-HSA-211999	8	CYP2C19, CYP2C8, CYP2C9	CYP2C19, CYP2C9
R5	Synthesis of (16-20)-hydroxyeicos-atetraenoic acids (HETE)	https://reactome.org/content/detail/R-HSA-2142816	8	CYP1A2, CYP2C19, CYP2C8, CYP2C9	CYP2C19, CYP2C9
R6	Synthesis of epoxy (EET) and dihydroxyeicosatrienoic acids (DHET)	https://reactome.org/content/detail/R-HSA-2142670	8	CYP1A2, CYP2C19, CYP2C8, CYP2C9	CYP2C19, CYP2C9
R7	Prednisone ADME	https://reactome.org/content/detail/R-HSA-9757110	10	ALB, CYP3A4	ABCB1, CYP3A4
R8	Phase I - Functionalization of compounds	https://reactome.org/content/detail/R-HSA-211945	13	CYP3A4	CYP3A4
R9	Xenobiotics	https://reactome.org/content/detail/R-HSA-211981	16	CYP2C18, CYP2C19, CYP2C8, CYP2C9, CYP3A4	CYP2C19, CYP2C9, CYP3A4, CYP3A5
R10	Aspirin ADME	https://reactome.org/content/detail/R-HSA-9749641	25	ALB, CYP2C19, CYP2C8, CYP2C9, CYP3A4	CYP2C19, CYP2C9, CYP3A4

Proteins in the hypotheses: family background (project text, not a statement about this pair)

CYP2C19 An enzyme of the cytochrome P450 family. These work mainly in the liver and gut and break down many medicines so the body can clear them.

CYP2C9 An enzyme of the cytochrome P450 family. These work mainly in the liver and gut and break down many medicines so the body can clear them.

CYP3A4 An enzyme of the cytochrome P450 family. These work mainly in the liver and gut and break down many medicines so the body can clear them.

5 BIO-GINE research analysis

CALIBRATED SCORE	RAW SCORE	REGIME	RELIABILITY FOR THIS PAIR
0.811	0.999972	S2	like the held-out pairs of the frozen test

Model and checkpoint	BIO-GINE M4 · bd45f84e3c1b2c33 · seed 0
Regime	S2 — one drug was outside the training set
Splits A / B	val / train
Pair's record seen in training	no
Protein data available to the model	A: yes · B: yes

How to read the score

The BIO-GINE score says how closely a pair resembles pairs already recorded as interacting. Calibration was fitted on validation pairs with equal shares of documented and randomly drawn pairs, so the number is not a probability of harm, not a frequency of interactions in practice, and not a severity. Its use is ranking for research.

Uncertainty

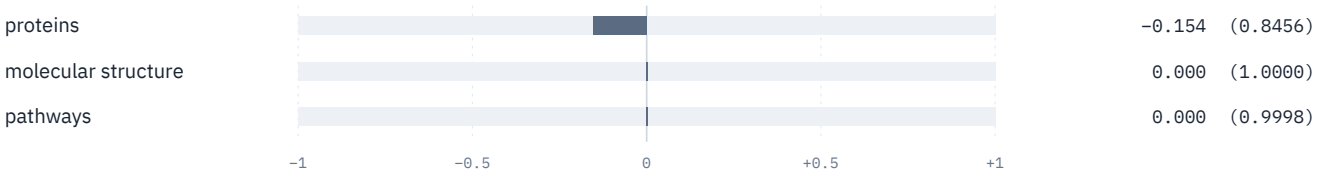
The service uses one model checkpoint (seed 0) and provides neither a per-pair confidence interval nor an ensemble estimate. Measured performance for this regime is given instead — an average over many pairs, not confidence in this pair.

Measured performance for this regime

METRIC	MEAN ± SD	SEEDS	EVALUATION VIEW	SOURCE
AUPRC	0.812 ± 0.010	5	pooled drug-disjoint test (S2 and S3; 91% S2)	paper/FACTS.json models.M4_pooled (v2_final_test.csv @ frozen tag)
AUROC	0.818 ± 0.010	5	pooled, both annotated	reports/v3_coverage_reliability.json

What the model leans on (one kind of data zeroed)

Change in the raw score when one kind of data is removed for both drugs. A minus sign means the score drops without that data. In brackets: the raw score without this data.



Raw score: 0.999972

The raw score is near the edge of the scale (0 or 1); changes are compressed there, and reliance can look smaller than it is.

This shows which data the model leans on for this pair, not why the drugs interact: the model never saw zeroed inputs in training.

Method check: reproduces the frozen V2 analysis on 84,690 test rows, max |Δ| = 6.1e-6.

Method: branch zeroed before the frozen fusion layer for both drugs; mask unchanged (as scripts/49_v2_interpretability.py at the frozen tag)

Notes from the service

- One drug was in the validation set: the model was not trained on it, but it was used to choose the configuration and the calibration.
- The calibrated score was fitted on validation pairs with equal numbers of documented and randomly drawn pairs. It is not a probability of harm, nor how often interactions happen in practice.

6 Structured explanation from the service

The text below was written by the service with deterministic templates over the retrieved records (no language model) and checked by it for advice wording. Bracketed labels point to records: [n] evidence, H hypothesis, M model, R Reactome pathway.

What is known

Source — DrugBank describes this combination as: “The serum concentration of one of the two drugs can be increased when it is combined with the other drug.” [1]

Caveat — For this record type the drug order in the DrugBank export proved unreliable (we measured it), so the text does not say which drug affects which. [1]

Why this may happen

Hypothesis — fluconazole is recorded as inhibiting CYP2C19, and warfarin as something CYP2C19 processes. With CYP2C19 inhibited, warfarin may be broken down more slowly than usual. H1

Hypothesis — fluconazole is recorded as inhibiting CYP2C9, and warfarin as something CYP2C9 processes. With CYP2C9 inhibited, warfarin may be broken down more slowly than usual. H2

Hypothesis — fluconazole is recorded as inhibiting CYP3A4, and warfarin as something CYP3A4 processes. With CYP3A4 inhibited, warfarin may be broken down more slowly than usual. H3

Caveat — These hypotheses agree with a DrugBank interaction record. That is not independent confirmation: DrugBank may have derived the record from the same protein annotations. H1 H2 H3 [1]

Caveat — The databases record neither how strongly a drug inhibits or induces a protein nor how much of a drug's handling that protein does. These are hypotheses, not a conclusion that an interaction will happen. H1 H2 H3

Annotation — Proteins of both drugs take part in 10 shared biological process(es) in Reactome — for example “Biosynthesis of maresin-like SPMs”. A shared process is background, not a cause of interaction. R

What the AI found

Model — BIO-GINE finds this pair strongly resembles pairs already recorded as interacting. This is resemblance to recorded pairs, not an estimate of harm. M

Model — For this pair the model's score depends most on the protein data: without it the score moves by 0.15. This says what the model leans on, not why the drugs interact. M

Caveat — One drug was in the validation set: the model was not trained on it, but it was used to choose the configuration and the calibration. M

Caveat — The calibrated score was fitted on validation pairs with equal numbers of documented and randomly drawn pairs. It is not a probability of harm, nor how often interactions happen in practice. M

How certain is this?

Source — Documented: 1 source record(s). This is the most reliable part of the page. [1]

Caveat — No severity is assigned: the DrugBank export has none, and for labels only the section that names the drug is shown.

Hypothesis — The 3 mechanism hypothesis(es) are assembled from each drug's documented roles separately. None was tested for this pair. H1 H2 H3

Model — Model quality on held-out pairs where the model did not see one of the drugs in training (as here): AUPRC 0.812 ± 0.010 over 5 runs. That is an average over many pairs, not confidence in this particular pair. M

What is missing

Missing — The source “FDA labels (openFDA)” could not be checked right now. Its data is not part of this answer.

Caveat — Doses, duration, age, kidney and liver function, other medicines and individual factors are not taken into account — everything that decides whether an interaction shows up and how strongly.

Caveat — Decisions about medicines belong to a doctor or pharmacist. This page cannot replace them.

Method: deterministic templates over retrieved records; no language model

7 Sources and reproducibility

Evidence records

#	SOURCE	RECORD ID	RETRIEVED	URL
[1]	DrugBank DDI records via Therapeutics Data Commons · PyTDC 1.1.15, tdc.multi_pred.DDI(name='DrugBank')	DB00682|DB00196|Y=73	2026-08-24	https://tdcommons.ai/multi_pred_tasks/ddi/

Protein annotations the hypotheses rest on

DRUG	PROTEIN	ROLES	SOURCE	RETRIEVED	PUBMED
Fluconazole	CYP2C19 P33261	inhibitor raw action: inhibitor	DrugBank drug-protein annotation · 5.1 (export 2025-01-02)	2026-08-28	16141567, 10709776
Warfarin	CYP2C19 P33261	inhibitor, substrate raw action: substrate|inhibitor	DrugBank drug-protein annotation · 5.1 (export 2025-01-02)	2026-08-28	19515014, 16162970, 16372822

DRUG	PROTEIN	ROLES	SOURCE	RETRIEVED	PUBMED
Fluconazole	CYP2C9 P11712	inhibitor raw action: inhibitor	DrugBank drug-protein annotation · 5.1 (export 2025-01-02)	2026-08-28	16141567, 8801056, 10952477, 11475469, 10709776
Warfarin	CYP2C9 P11712	inducer, substrate raw action: substrate|inducer	DrugBank drug-protein annotation · 5.1 (export 2025-01-02)	2026-08-28	29283396, 16372822, 20735727
Fluconazole	CYP3A4 P08684	inhibitor raw action: inhibitor	DrugBank drug-protein annotation · 5.1 (export 2025-01-02)	2026-08-28	16141567, 8801056, 15969931
Warfarin	CYP3A4 P08684	inducer, substrate raw action: substrate|inducer	DrugBank drug-protein annotation · 5.1 (export 2025-01-02)	2026-08-28	19515014, 16372822, 20735727

PubMed: https://pubmed.ncbi.nlm.nih.gov/<PMID>/ · UniProt: https://www.uniprot.org/uniprotkb/<accession>/entry

Provenance and parameters

Endpoint (report)	`http://127.0.0.1:8765/api/v3/report`
Request body (report)	`{"drug_a":"DB00682","drug_b":"DB00196","include_labels":true}`
Response received by the browser (report)	2026-10-15T04:51:44.755Z
Report generated by the service (generated_at)	2026-10-15T04:51:44+00:00
API report version	3.0
Explanation method	deterministic templates over retrieved records; no language model
Model	BIO-GINE M4 · bd45f84e3c1b2c33 · seed 0
Frozen research tag	`v2-final-github-safe-2026-09-03 · 92c481eeaba8faff991ced850e1c4de418ea31b0`
Site	http://127.0.0.1:4173
Site build commit	not recorded
PDF generator	DDI-Net web, in-browser export (@react-pdf/renderer)
Generated	15 October 2026, 04:51:45 (UTC, UTC+00:00)
Generated (UTC)	2026-10-15T04:51:45.099Z
Language	English
SHA-256 of the response data	`80ddd4eaafa0c097ddbb14fbcdeefa82107b0444a050233783ecd33cd91efdc3` Hash of the service response(s) as canonical JSON (keys sorted, serialised with JSON.stringify). The same hash is in the companion JSON file, available from the “JSON” button next to “Export PDF”.

Retrieval times are given per record (retrieved_at) as returned by the service: the export date for local extracts, the request date for labels.

8 Limitations

Core limitations

- DDI-Net is a research tool, not clinical decision support and not a medical device. Decisions about medicines belong to a doctor or pharmacist.
- Unlabelled pairs are not confirmed safe. A missing record means only that the sources checked do not contain one.
- BIO-GINE scores are not calibrated probabilities of clinical harm: temperature scaling was fitted on validation pairs at 50% prevalence against randomly drawn unlabelled pairs.
- The drug order in the DrugBank (TDC) export was verified by measurement for only two interaction types; for the others it is either meaningless (symmetric types) or unverified, so directional claims (“A raises the level of B”) are not made.
- External medical evidence (FDA labels, RxNorm) may be incomplete or unavailable at query time; its status is listed under “Sources checked”.

Limitations reported by the service for this report

- Only the 1,705 drugs of the research dataset; 1,497 of them have an international name.
- DrugBank records carry no severity and do not state their basis. For many record types the drug order in the export is unreliable (measured).
- Labels are searched by drug name. Statements about whole drug classes (“strong CYP3A4 inhibitors”) are not recognised, so no match does not mean the label is silent.
- Mechanism hypotheses are assembled from each drug's roles separately; strength of effect and the protein's share of the work are unknown.
- The BIO-GINE score is a research signal of resemblance to documented pairs. It is not a probability of harm.

> RESEARCH USE ONLY
> This is a research prototype, not a medical device. It does not replace a doctor or pharmacist and cannot say whether a combination is safe for a particular person.

A Appendix A. Machine-readable summary (JSON)

Key fields as JSON for automated checking. The full service responses are in the companion JSON file (same SHA-256).


```
{
  "schema": "ddinet.pdf-summary/1",
  "report_id": "DDI-20261015-045145-80DDD4EA",
  "generated_at": "2026-10-15T04:51:45+00:00",
  "generated_at_utc": "2026-10-15T04:51:45.099Z",
  "time_zone": "UTC",
  "language": "en",
  "payload_sha256": "80ddd4eaafa0c097ddbb14fbcdeefa82107b0444a050233783ecd33cd91efdc3",
  "inputs": [
    {"endpoint":"http://127.0.0.1:8765/api/v3/report","request":{"drug_a":"DB00682","drug_b":"DB00196","include_labels":true},
      "received_at":"2026-10-15T04:51:44.755Z"}
  ],
  "server_generated_at": "2026-10-15T04:51:44+00:00",
  "report_version": "3.0",
  "drugs": [
    {"role":"A","id":"DB00682","inn":"warfarin","name_ru":"варфарин"},
    {"role":"B","id":"DB00196","inn":"fluconazole","name_ru":"флуконазол"}
  ],
  "evidence_status": "documented_no_severity",
  "evidence": [
    {"ref":"[1]","id":"drugbank-tdc:DB00682>DB00196:t73","category":"database_interaction_record","source":
      "DrugBank DDI records via Therapeutics Data Commons","record_id":"DB00682|DB00196|Y=73","direction":"not_verified",
      "retrieved_at":"2026-08-24","url":"https://tdcommons.ai/multi_pred_tasks/ddi/"}
  ],
  "conflicts": 0,
  "hypotheses": [
    {"ref":"H1","gene":"CYP2C19","uniprot":"P33261","relation":"enzyme","pattern":"inhibitor_substrate","actor":"DB00196",
      "affected":"DB00682"},
    {"ref":"H2","gene":"CYP2C9","uniprot":"P11712","relation":"enzyme","pattern":"inhibitor_substrate","actor":"DB00196","affected":
      "DB00682"},
    {"ref":"H3","gene":"CYP3A4","uniprot":"P08684","relation":"enzyme","pattern":"inhibitor_substrate","actor":"DB00196","affected":
      "DB00682"}
  ],
  "shared_pathways_total": 10,
  "model": {
    "model": "BIO-GINE M4",
    "checkpoint": "bd45f84e3c1b2c33",
    "available": true,
    "calibrated_score": 0.8108211344965719,
    "raw_score": 0.9999718803394634,
    "regime": "S2",
    "reliability": "held_out_like",
    "abstain": false
  },
  "sources_checked": [
    {"source":"drugbank_tdc","status":"ok"},
    {"source":"openfda_label","status":"unavailable"}
  ]
}
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